

MILESTONE 1

1. Dataset Description

1.1 Dataset Overview

- The project uses the **VinBigData Chest X-ray Abnormalities Detection Dataset**
- This is a large-scale medical imaging dataset designed for object detection of thoracic abnormalities in chest radiographs.

The dataset contains:

- **15,000+ chest X-ray images**
- Images stored in **DICOM (.dicom / .dcm) format**
- **14 abnormality classes**
- One additional class: **“No Finding”**
- Bounding box annotations provided in **CSV format**
- Annotations labeled by **multiple expert radiologists**

1.2 Dataset Structure

The dataset consists of:

- train/ → DICOM images
- test/ → DICOM images
- train.csv → Bounding box annotations
- sample_submission.csv

Each row in the CSV contains:

- image_id
- class_name
- class_id
- x_min, y_min, x_max, y_max
- Radiologist ID

For “No Finding” cases, bounding box coordinates are **NaN**.

1.3 Classes in the Dataset

The dataset includes 14 disease categories such as:

- Atelectasis
- Cardiomegaly

- Pleural Effusion
- Pneumothorax
- Lung Opacity
- Consolidation
- Calcification
- Nodule/Mass
- ILD
- Aortic Enlargement
- Pleural Thickening
- Pulmonary Fibrosis
- Infiltration
- Other Lesion

Plus:

- **No Finding**

2. Data Preprocessing Steps

Data preprocessing is essential in medical imaging to ensure model stability, accuracy, and consistency.

The following preprocessing pipeline was implemented.

Step 1: DICOM to PNG Conversion

Medical images were originally stored in DICOM format.

Why conversion was required:

- Deep learning frameworks (YOLO, PyTorch) commonly use PNG/JPEG
- Simplifies visualization
- Reduces complexity of data loading

Process:

- Read DICOM using pydicom
- Extract pixel array
- Normalize pixel intensities
- Convert to 8-bit format
- Save as PNG

Only the first 1000 images were converted initially to reduce memory usage.

Step 2: Annotation Parsing & Conversion to YOLO Format

Original annotations were provided in CSV format with bounding box coordinates:

- x_min, y_min, x_max, y_max

These were converted into YOLO format:

- class_id x_center y_center width height

Where:

- Coordinates are normalized between 0 and 1
- Calculated relative to image width and height
- “No Finding” cases were skipped because they contain no bounding box.
- Each image received a corresponding .txt file.

Step 3: Bounding Box Verification

To verify correctness:

- Bounding boxes were drawn on sample images
- YOLO normalized values were converted back to pixel coordinates
- Rectangles were visualized using OpenCV

This confirmed successful annotation conversion.

Step 4: Grayscale Conversion

Chest X-rays are inherently grayscale.

Images were converted to single-channel grayscale to:

- Reduce computational complexity
- Improve model focus on intensity patterns
- Remove unnecessary color channels

Step 5: Border Cropping

Black borders around X-ray images were removed to:

- Reduce irrelevant background
- Focus on thoracic region
- Improve detection performance

Thresholding and bounding rectangle extraction were used.

Step 6: Image Resizing

All images were resized to:

- 640×640

Reason:

- Required input size for YOLO
- Ensures uniformity
- Improves batch processing efficiency

Step 7: Contrast Enhancement (CLAHE)

CLAHE (Contrast Limited Adaptive Histogram Equalization) was applied.

Purpose:

- Enhance subtle abnormalities
- Improve local contrast
- Prevent over-amplification of noise

This is especially useful in medical imaging.

Step 8: Light Denoising

Gaussian blur was applied to:

- Reduce minor noise
- Smooth small artifacts
- Improve visual clarity

Kernel size used: (3,3)

Step 9: Processed Image Verification

Processed images were visualized to confirm:

- Correct size (640×640)
- Proper contrast
- Successful border removal
- Image clarity

Step 10: Train / Validation / Test Split

The dataset was divided into:

- 70% Training
- 20% Validation
- 10% Testing

Using `train_test_split` with a fixed random seed for reproducibility.

Directory structure created:

yolo_dataset/

images/

- train/
- val/
- test/

labels/

- train/
- val/
- test/

Final Outcome of Milestone-1

- ✓ Dataset downloaded and verified
- ✓ DICOM images converted to PNG
- ✓ Annotation converted to YOLO format
- ✓ Bounding boxes validated visually
- ✓ Preprocessing pipeline implemented
- ✓ Images resized and enhanced
- ✓ Dataset split into train/val/test
- ✓ YOLO-compatible folder structure created